

Mathematical and Logical Foundations of Computer Science

Lecture 6 - Propositional Logic (Classical Reasoning)

Vincent Rahli

(some slides were adapted from Rajesh Chitnis' slides)

University of Birmingham

Where are we?

- ▶ Symbolic logic
- ▶ **Propositional logic**
- ▶ Predicate logic

Today

- ▶ Classical Reasoning
- ▶ Constructive vs. Classical Natural Deduction

Further reading

- ▶ Chapter 5 of
http://leanprover.github.io/logic_and_proof/

Recap: Propositional logic syntax

Syntax:

$$P ::= a \mid P \wedge P \mid P \vee P \mid P \rightarrow P \mid \neg P$$

Two special atoms:

- ▶ $\top$ which stands for True
- ▶ $\perp$ which stands for False

We also introduced four connectives:

- ▶ $P \wedge Q$: we have a proof of both P and Q
- ▶ $P \vee Q$: we have a proof of at least one of P and Q
- ▶ $P \rightarrow Q$: if we have a proof of P then we have a proof of Q
- ▶ $\neg P$: stands for $P \rightarrow \perp$

Recap: Proofs

Natural Deduction

introduction/elimination rules

natural proofs

$$\frac{\begin{array}{c} \neg \quad 1 \\ A \\ \vdots \\ B \end{array}}{A \rightarrow B} \quad 1 \ [\rightarrow I]$$

Classical Reasoning

The proof systems we have seen so far are sometimes called **constructive** or **intuitionistic**, i.e., **proofs** can be viewed as **programs**:

- ▶ A proof of $A \wedge B$ can be viewed as a **pair** of a proof of A and a proof of B
- ▶ A proof of $A \rightarrow B$ can be viewed as a **procedure** which transforms evidence for A into evidence for B
- ▶ A proof of $A \vee B$ is either a proof of A or a proof of B , which indicates which one it is

There are other proof systems, called **classical**, which

- ▶ rely on Boolean truth values
- ▶ introduce additional reasoning principles

Classical Reasoning: Proof by Contradiction

A typical classical reasoning principal is the “**proof by contradiction**” proof technique

Example: Euclid's proof of infinitude of primes

- ▶ **Assume the negation:** Suppose there are only finitely many primes, say $p_1, p_2, \dots, p_r$
- ▶ Consider the number $n = (p_1 \times p_2 \times \dots \times p_r) + 1$
- ▶ Then n cannot be a prime (by assumption)
- ▶ But none of the primes $p_1, p_2, \dots, p_r$ can divide n
- ▶ **Contradiction**

Proof by Contradiction:

- ▶ If $\neg A \rightarrow \perp$ then A
- ▶ That is, $\neg\neg A \vdash A$

Negation of a negation is?

Can we deduce A and $\neg\neg A$ from each other? That is, are they equivalent?

One direction is easy: $A \vdash \neg\neg A$

Here is the proof:

$$\frac{\frac{A \quad \overline{\neg A}^1}{\perp} [\neg E]}{\neg\neg A}^1 [\neg I]$$

Can we show the other direction, i.e., $\neg\neg A \vdash A$?

Not using the current set of inference rules we have!

Classical vs. Intuitionistic Reasoning in Natural Deduction

Two more (equivalent) assumptions/rules

Law of Excluded Middle (LEM)

- ▶ For each A , we can always prove one of A or $\neg A$
- ▶ i.e., $\vdash A \vee \neg A$
- ▶ E.g., we can assume every even natural number > 2 is the sum of two primes, or not, without knowing which one is true

Double Negation Elimination (DNE)

- ▶ $\neg\neg A \vdash A$
- ▶ Equivalently, $(\neg A) \rightarrow \perp \vdash A$
- ▶ “proof by contradiction”

Classical vs. Intuitionistic Reasoning in Natural Deduction

Two more (equivalent) assumptions/rules

As rules:

$$\frac{}{A \vee \neg A} \quad [LEM] \qquad \frac{\neg \neg A}{A} \quad [DNE]$$

Classical reasoning allows using these two rules

We so far have not used them, and were therefore using what is called **constructive** or **intuitionistic** logic

LEM implies DNE

Assuming $A \vee \neg A$, infer $\neg\neg A \vdash A$

Here is a proof:

$$\frac{A \vee \neg A \quad \frac{\frac{\overline{A}^1}{A \rightarrow A} \text{ } 1 [\rightarrow I] \quad \frac{\frac{\frac{\overline{\neg A}^2}{\neg\neg A} \text{ } [\neg E] \quad \frac{\perp}{A} [\perp E]}{\neg A \rightarrow A} \text{ } 2 [\rightarrow I]}{A} \text{ } [\vee E]}{A}$$

DNE implies LEM

Assuming $\neg\neg A \vdash A$, infer $\vdash A \vee \neg A$

Here is a proof:

$$\frac{\frac{\frac{\overline{\neg(A \vee \neg A)}}{1} \quad \frac{\frac{\overline{A}}{2} \quad [\vee I_L]}{A \vee \neg A} \quad [\neg E]}{\perp} \quad [\neg I]$$
$$\frac{\frac{\overline{\neg(A \vee \neg A)}}{1} \quad \frac{\frac{\overline{\neg A}}{2} \quad [\neg I]}{A \vee \neg A} \quad [\vee I_R] \quad [\neg E]}{\perp} \quad [\neg I]$$
$$\frac{\frac{\overline{\neg\neg(A \vee \neg A)}}{1} \quad [\neg I]}{A \vee \neg A} \quad [DNE]$$

Contrapositive

Given an implication $A \rightarrow B$, the formula $\neg B \rightarrow \neg A$ is called the “**contrapositive**”

Can we prove that an implication implies its contrapositive?

$$A \rightarrow B \vdash \neg B \rightarrow \neg A$$

Here is a proof (intuitionistic):

$$\frac{\frac{A \rightarrow B \quad \overline{A}^2}{B} \quad [\rightarrow E] \quad \frac{}{\neg B}^1}{\perp} [\neg E] \quad \frac{}{\neg A}^2 [\neg I] \quad \frac{}{\neg B \rightarrow \neg A}^1 [\rightarrow I]$$

The other direction holds in classical logic (next slide)

Contrapositive

Given an implication $A \rightarrow B$, the formula $\neg B \rightarrow \neg A$ is called the “**contrapositive**”

Can we prove that an implication follows from its contrapositive?
 $\neg B \rightarrow \neg A \vdash A \rightarrow B$

Here is a proof (classical):

$$\frac{\frac{\frac{\overline{A} \quad 1 \quad \frac{\frac{\neg B \rightarrow \neg A \quad \overline{\neg B} \quad 2}{\neg A} [\rightarrow E]}{\perp} [\neg E]}{\frac{\perp}{\neg\neg B} \quad 2 [\neg I]} [\text{DNE}]}{\frac{B}{A \rightarrow B} \quad 1 [\rightarrow I]}$$

We used DNE, and hence this proof uses classical reasoning!

Classical Reasoning Through Examples

We will present classical proofs of:

- ▶ $(A \rightarrow B) \vee (B \rightarrow A)$
- ▶ $(\neg B \rightarrow \neg A) \rightarrow (A \rightarrow B)$

We saw a classical proof of $(\neg B \rightarrow \neg A) \rightarrow (A \rightarrow B)$ before using DNE – we will present an alternative proof that uses LEM instead

Which we will prove in classical Natural Deduction.

Example 1

Provide a classical Natural Deduction proof of $(A \rightarrow B) \vee (B \rightarrow A)$

$$\begin{array}{c}
\frac{\overline{A \vee \neg A} \quad [LEM] \quad \frac{\frac{\frac{\frac{\overline{A} \quad 1}{B \rightarrow A} \quad 2 \quad [\rightarrow I]}{(A \rightarrow B) \vee (B \rightarrow A)} \quad [\vee I_R] \quad \frac{\overline{A \rightarrow (A \rightarrow B) \vee (B \rightarrow A)} \quad 1 \quad [\rightarrow I]}{\overline{A \vee \neg A} \quad [LEM] \quad \frac{\frac{\frac{\frac{\overline{\neg A} \quad 3 \quad \overline{A} \quad 4}{\perp} \quad [\perp E]}{A \rightarrow B} \quad 4 \quad [\rightarrow I]}{(A \rightarrow B) \vee (B \rightarrow A)} \quad [\vee I_L] \quad \frac{\overline{\neg A \rightarrow (A \rightarrow B) \vee (B \rightarrow A)} \quad 3 \quad [\rightarrow I]}{\overline{(A \rightarrow B) \vee (B \rightarrow A)} \quad [\vee E]}
\end{array}$$

Hypotheses:

- ▶ hyp. 1: A
- ▶ hyp. 2: B
- ▶ hyp. 3: $\neg A$
- ▶ hyp. 4: A

Example 2

Provide a classical Natural Deduction proof of

$$(\neg B \rightarrow \neg A) \rightarrow (A \rightarrow B)$$

Here is a proof:

$$\begin{array}{c}
 \frac{\frac{\frac{\frac{}{\neg B \rightarrow \neg A} 1}{\neg B} 4}{\neg A} [\rightarrow E] \quad \frac{}{A} 2}{\neg A} [\neg E] \\
 \frac{\frac{}{B \vee \neg B} [LEM] \quad \frac{\frac{}{B} 3}{B \rightarrow B} 3 [\rightarrow I] \quad \frac{\frac{\frac{}{\bot} [\bot E]}{B} [\perp E] \quad \frac{}{\neg B \rightarrow B} 4 [\rightarrow I]}{B} [\vee E]}{\frac{\frac{}{A \rightarrow B} 2 [\rightarrow I]}{(\neg B \rightarrow \neg A) \rightarrow (A \rightarrow B)} 1 [\rightarrow I]}
 \end{array}$$

Hypotheses:

- ▶ hyp. 1: $\neg B \rightarrow \neg A$
- ▶ hyp. 2: A
- ▶ hyp. 3: B
- ▶ hyp. 4: $\neg B$

Conclusion

What did we cover today?

- ▶ Classical Reasoning
- ▶ Constructive vs. Classical Natural Deduction

Further reading

- ▶ Chapter 5 of http://leanprover.github.io/logic_and_proof/
- ▶ “Proofs and Types”, Girard, Taylor, and Lafont, Chapter 5

Next time

- ▶ Propositional logic's (classical) semantics